

POLARIZED LIGHT

CONVENTIONAL DESCRIPTION OF POLARIZED LIGHT

1.1. Introduction. In this book the term *light* stands for *electromagnetic radiation*. Usually we have in mind the 400–700 m μ range (visual range), but often we include also the shorter-wavelength range (ultraviolet) and the longer-wavelength range (infrared). On some occasions we include also the x-ray and gamma-ray ranges and the radio range. The total range in which polarization plays a part covers more than sixty octaves.

Polarized light is one of nature's ultimates. A slender, monochromatic, *polarized* ray cannot be subdivided into simpler components: no simpler components exist. The process of analysis can advance no further.

There is much to be gained, however, by considering how a beam of polarized light behaves and how it may be depicted. When such a beam encounters a birefringent crystal, a dichroic film, or an oblique dielectric surface, a great variety of behaviors may result. The question is: Can we find, for the polarized beam, a representation so pertinent and so versatile that, merely by examining the representation, we can predict the outcome of any given encounter?

Fortunately, several highly successful representations have been invented. Some are pictorial, others mathematical. Some are well suited to solving simple problems, others are to be preferred when the problems are complicated.

To ask whether a given representation is "true" is futile. It must suffice that the representation assists ready recollection of the behavior and permits easy solution of the various problems encountered.

The present chapter reviews the classical (pictorial and wave-train) methods of representing polarized light. Chapter 2 considers certain more modern and more powerful methods.

Polarized light, besides being of interest *per se*, serves as a tool, or probe, for evaluating the properties of matter. The tool exhibits the ultimate in speed, and perhaps the ultimate in delicacy and convenience. It has the merit of being completely convertible; that is, the polarization form can be altered at will, with no loss in power and no increase in entropy flux. In many respects, polarized light, being the simplest kind of light, is easier to deal with than ordinary light: the physical manipulations may be cleaner, and the mathematical procedures for predicting the experimental outcomes are simpler. Physicists and chemists find that polarized light has uses far beyond those of unpolarized light. Biologists, astronomers, and engineers find that polarized light solves many problems that are otherwise insoluble. If light is man's most useful tool, polarized light is the quintessence of utility.

In preparing this book the author faced a major problem as to conventions. The crux of the problem was the large number of branches of optics that must be brought into one consistent family. Traditionally, users of saccharimeters and other polarimeters employ a certain set of sign conventions, persons dealing with dichroism employ certain conventions, and similarly for persons dealing with crystallography, wave theory, the Stokes vector, the Poincaré sphere, the Mueller calculus, and the Jones calculus. Ordinarily, the incompatibility of the various sets of conventions as to signs, handedness, etc., is unnoticed and unimportant. In this book, however, one universally self-consistent set of conventions is mandatory. Accordingly, some conflict with various lesser sets is unavoidable.

1.2. Classical Pictorial Specification of a Polarized Wave Train. The classical description of a polarized wave train is well known (see, for example, Ditchburn, D-10, and Jenkins and White, J-9), and needs only brief review here.

From the standpoint of classical physics, light consists of electromagnetic waves whose vibrations are transverse to the propagation direction. *Polarized* light is light whose vibration pattern exhibits *preference*: preference as to transverse direction, or preference as to the handedness associated therewith. Different kinds of preference are

indicated in Fig. 1.1. Each drawing, called a snapshot pattern, describes the monochromatic wave train at a single instant in time; the curve may be thought of as a smooth line joining the tips of a large number of vectors that indicate the directions and magnitudes of the *electric* field at various positions along the center line of the beam. The convention with respect to right and left circular polarization is easily remembered: right circular polarization is portrayed by means of a right-handed helix, such as the thread of a typical machine screw.

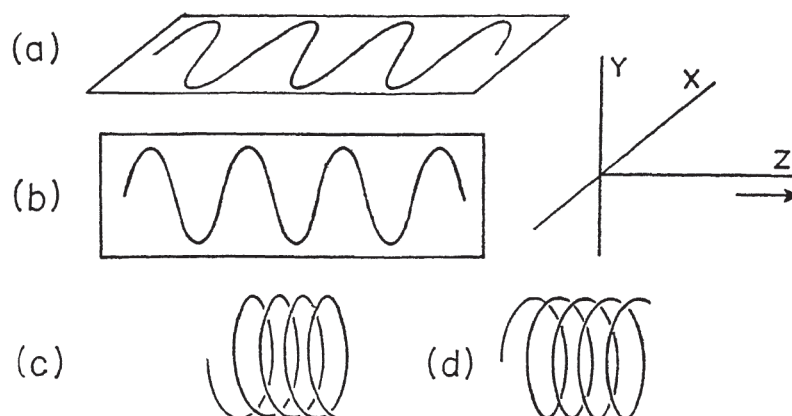


FIG. 1.1. Snapshot patterns of a horizontally traveling beam of monochromatic light that is polarized (a) horizontally, (b) vertically, (c) right circularly, and (d) left circularly.

(The reader will recall that a right-handed helix continues to appear right-handed no matter what the observer's viewpoint; consequently the present definition is free from ambiguity.) The pattern may be drawn with respect to a right-handed set of cartesian coordinates, Z being the direction of propagation, and X and Y being horizontal.

Workers in different fields (such as crystallography, theoretical physics, saccharimetry, radio technology) may employ conflicting definitions. The definitions used in this book are believed to represent the best compromise. Care has been taken to word the definitions clearly and to use them consistently.

One could, of course, deal with the magnetic, rather than the electric, vibration. When light is traveling in a vacuum or other isotropic medium, these two vibrations are orthogonal (perpendicular) and their magnitudes are always proportional to one another. To specify

one is tantamount to specifying both. The decision to concentrate on the *electric* vibration is conventional, and pays tribute to the dominant role of the electric vector in the more familiar absorption processes.

The *sectional pattern* (Fig. 1.2) is perhaps the most familiar of all the characterizations. A horizontally polarized beam is portrayed as a

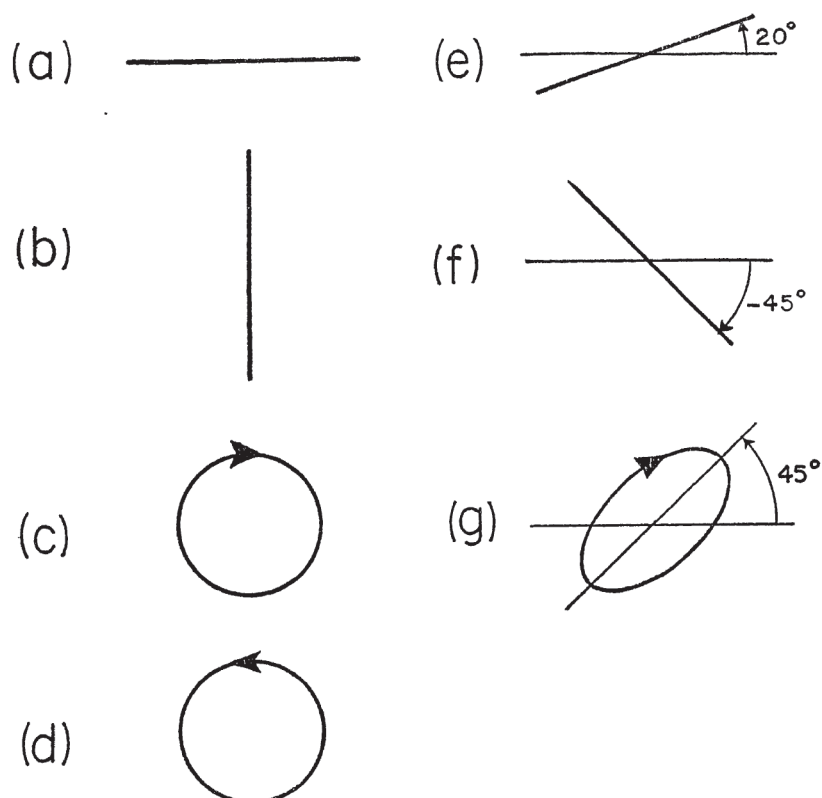


FIG. 1.2. Sectional pattern of a beam polarized (a) horizontally, (b) vertically, (c) right circularly, (d) left circularly, (e) linearly at 20° , (f) linearly at -45° , (g) right elliptically at 45° .

short horizontal line; vertical polarization is indicated by a vertical line. Right-circular polarization is portrayed by a circle having a clockwise sense. The sectional pattern may be thought of as an end view of the snapshot pattern, as seen by an observer who is situated in the path of the beam (specifically, far out on the Z -axis) and is looking toward the light source, which is at the origin of coordinates.

The clockwise sense of the circle describing right-circular polarization is consistent with the definition involving a *right-handed helix*: if a right-handed helix is moved bodily toward an observer (without rotation) through a fixed, transverse, reference plane, the point of intersection of helix and plane executes a *clockwise* circle.

The sectional patterns are easy to draw, even for light that is polarized elliptically. Also, they can embrace polychromatic light; however, we must then think of the scale of the pattern as changing more or less rapidly, depending on the frequency bandwidth of the beam; consequently the patterns (Fig. 1.3) cease to be of simple, closed type.

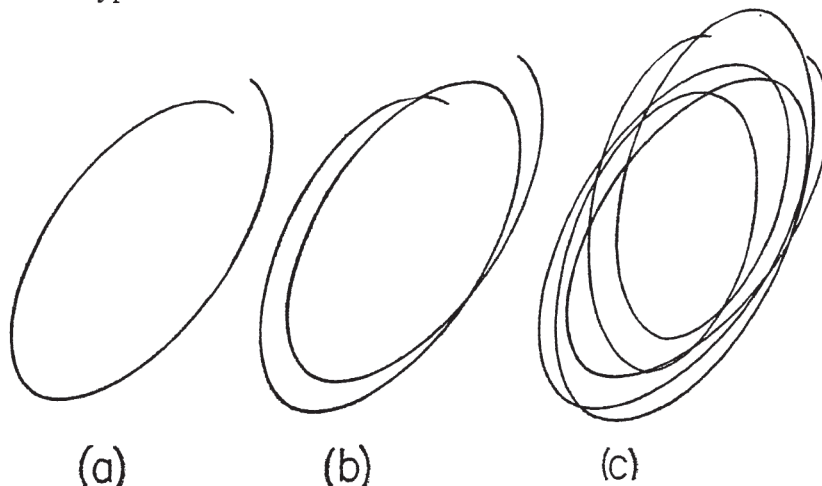


FIG. 1.3. Appearance of sectional pattern of an elliptically polarized beam having small but appreciable bandwidth, assuming observation times of (a) about 1 cycle, (b) about 2 cycles, (c) many cycles.

The general sectional pattern of a monochromatic beam — an ellipse — may be described with the aid of the terms defined in Fig. 1.4. The angle α (between the major semiaxis and the X -axis) is called the *azimuth* of the sectional pattern; $90^\circ \geq \alpha \geq -90^\circ$. The ratio b/a of the semiaxes is called the *ellipticity*; the symbol β may be used to represent $\arctan b/a$; $90^\circ \geq \beta \geq -90^\circ$. Ellipticity is used in preference to the eccentricity, which is defined as $(a^2 - b^2)^{1/2}/a$.

In some instances the ratio A_y/A_x is of interest; A_y is the maximum value of the Y -component of the electric vector, and A_x is the maximum value of the X -component. The angle $|\arctan (A_y/A_x)|$ will be

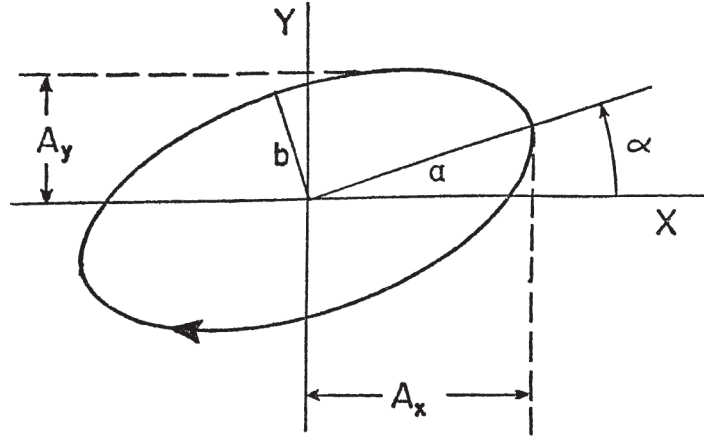


FIG. 1.4. Elliptically polarized light. In this example, $\alpha = 20^\circ$, the ellipticity $b/a = 0.4$, and the handedness is clockwise, as judged by an observer situated far out on the Z -axis and looking backward toward the source, which is at the origin.

called R . When α is in the neighborhood of $\pm 45^\circ$, or when the ellipse is very slender, the angles $|\alpha|$ and R are not very different, but under other circumstances they differ greatly.

Polarization Types and Forms. Linear polarization, circular polarization, and elliptical polarization may be referred to as the three *polarization types*. Obviously, the elliptical type includes the others as special cases; ellipticities of 0 and 1 correspond to linear and circular polarization respectively.

The linear type of polarization includes an infinite number of *polarization forms*, differing as to azimuth α . Circular polarization includes two forms, differing as to handedness. Elliptical polarization includes an infinite number of forms, differing as to azimuth, ellipticity, and handedness.

Orthogonal Forms. Two forms of linear polarization that differ by exactly 90° in azimuth are said to be orthogonal, assuming the directions of propagation to be the same (Fig. 1.5). Right- and left-circularly polarized beams are orthogonal. Two elliptically polarized beams are orthogonal if the azimuths of the major axes differ by 90° , the handednesses are opposite, and the ellipticities are identical.

Plane of Polarization. The expression *plane of polarization*, used by many authors, may be ambiguous. To some authors it means the plane containing the directions of propagation and of the electric

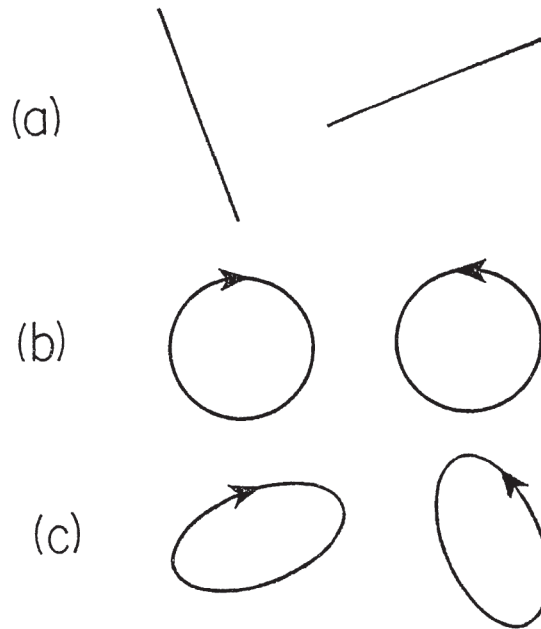


FIG. 1.5. Orthogonal pairs of beams polarized (a) linearly, (b) circularly, (c) elliptically.

vibration, while to others it means the plane containing the directions of propagation and of the *magnetic* vibration. Another drawback to the expression is that an experimenter can easily produce a number of beams that have the *same* plane of polarization yet *different* directions of electric vibration (Fig. 1.6). Likewise he can produce beams having *different* planes of polarization and the *same* direction of vibration.

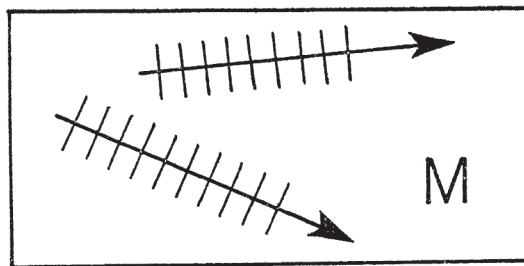


FIG. 1.6. Two beams having the same plane of polarization (plane *M*) yet different directions of electric vibration (indicated by hatch marks, all of which lie in plane *M*).

In this book these difficulties are avoided by the expedient of using terms descriptive of the key quantity, namely, the direction of the electric vibration. We refer to *linearly* polarized light and to the *direction of vibration*. We avoid expressions such as *plane polarized light*, and *plane of polarization*.

1.3. Mathematical Specification of a Polarized Wave Train. The mathematical specification of a wave train is explained in the standard textbooks on electromagnetic theory (D-10; A-1). Simple, monochromatic, linearly polarized trains of plane waves are propagated by means of transverse displacements varying sinusoidally with time and with position along the propagation direction. The magnitude ξ of the electric displacement $\mathbf{E}$ may be described by an expression such as

$$\xi = \sin(\omega t - 2\pi Z/\lambda),$$

where Z is the position along the axis of propagation, λ is the wavelength, ω is the angular frequency (2π times the ordinary frequency), and t is time.

To facilitate computations of certain sorts, one may introduce complex notation. To make the expressions more versatile, one may include a constant A , called the magnitude of the peak amplitude, and a quantity ϵ , called the epoch. The expression may take any of the following forms:

$$\begin{aligned}\xi &= A e^{i\epsilon} e^{i\omega t} e^{-i2\pi Z/\lambda} \\ &= A e^{i\epsilon} e^{i(\omega t - 2\pi Z/\lambda)} \\ &= A e^{i(\epsilon + \omega t - 2\pi Z/\lambda)} \\ &= A e^{i\phi},\end{aligned}$$

where $\phi = \epsilon + \omega t - 2\pi Z/\lambda$. The real part of ξ represents the instantaneous magnitude of the electric vector $\mathbf{E}$ (at time t and position Z). The quantity $A e^{i\phi}$ is called the complex amplitude. The quantity $\phi [= \epsilon + \omega t - 2\pi Z/\lambda]$, is the phase angle at time t and position Z . The intensity of the beam depends on A , and is, of course, proportional to A^2 .

The term *intensity* is used in this book in a number of ways. Sometimes it means the total power of the beam. On other occasions it means power per unit solid angle, or power per unit solid angle and per unit cross-sectional area. The intended meaning is usually made clear by the context. Formal definitions of intensity are presented by Chandrasekhar (C-8).

The direction of the electric vector does not appear in the equations, but may be specified separately — verbally or pictorially; or it may be specified by including unit vectors i and j , parallel to the X - and Y -axes respectively.

A circularly polarized beam of monochromatic light may be represented by a combination of two expressions, each having a complex magnitude of the form $Ae^{i\phi}$. One expression describes the vertical component (Y -component), and is written $A_y e^{i\phi_y}$; the other describes the horizontal component (X -component) and is written $A_x e^{i\phi_x}$. The amplitudes A_y and A_x are equal, and the phase angles ϕ_y and ϕ_x differ by 90° . If the difference $(\phi_y - \phi_x)$, called γ , is positive (90°), the light is *right*-circularly polarized. If $\gamma = -90^\circ$, the light is left-circularly polarized.

In the general case, the Y - and X -components differ in amplitude, and γ may have any value; the general sectional pattern is, of course, an ellipse. If $180^\circ > \gamma > 0^\circ$, the handedness is right; if $-180^\circ < \gamma < 0^\circ$, the handedness is left. When $\gamma = 0^\circ$, the pattern consists of a straight line (linear polarization), and when $|\gamma| = 90^\circ$ the pattern is a circle.

To predict the outcome of adding two monochromatic polarized beams, one adds their instantaneous electric vectors. In general, the two vectors have different directions in real, three-dimensional space, different frequencies, and unrelated phases; hence the result of the addition is a complicated and not very useful expression. In the simple case in which both beams are linearly polarized and have the same frequency and same phase, the procedure is simply to add the two vectors representing the root-mean-square electric vibrations of the two beams (Ref. W-1, p. 27).

If the two linearly polarized beams differ in phase (by some constant amount), the procedure is more complicated; if the phase difference is 180° , and if the two beams have equal intensity, the combined beam will have zero intensity. When two *coherent* beams intersect at a slight angle, the phase relation varies, of course, from one point to another in the region of intersection (as explained in Refs. D-10 and B-43, coherent beams are beams whose phases have a fixed, or virtually fixed, relation to one another); consequently the combined beams will have high intensity at some points and low intensity at others, so that an *interference pattern* results.

If the beams are completely *incoherent*, a short-cut procedure is

available: merely add the intensities of the beams. The sum of the intensities is the intensity of the combined beam.

Later chapters make it clear that the combining of beams can usually be handled more simply by certain modern methods than by the classical equations presented above. The Stokes vector provides an ideal basis for treating the combining of incoherent beams; the Jones vector is eminently applicable to coherent beams. These vectors are discussed in Chapter 2.

1.4. Unpolarized Light. Defined operationally, an unpolarized beam is a beam that, when operated on by any elementary kind of energy-conserving device that divides the beam into two completely polarized subbeams, yields subbeams that have *equal power* (in a time interval long enough to permit the powers to be measured). Thus if a beam is to qualify as unpolarized, it must exhibit no long-term preference as to lateral direction of vibration or as to handedness.

Can a perfectly monochromatic beam qualify as unpolarized? Obviously it cannot. Such a beam necessarily has a perfectly regular wave train, and consequently has a very definite and steady sectional pattern. Thus it exhibits polarization. Almost perfectly monochromatic radio waves are a common occurrence, and are found, of course, to exhibit a high degree of polarization.

Visible light, however, always possesses an appreciable bandwidth. Accordingly, such a beam may include many different forms of polarization simultaneously. If an experimenter is unable to detect any preponderant azimuth or handedness, he will perforce regard the beam as being unpolarized. (This subject has been explored by Langsdorf and DuBridge, L-14, and by Birge and DuBridge, B-29.) In this book the expression “monochromatic light” often appears; usually it means light that is *roughly* monochromatic and has sufficient bandwidth that unpolarized behavior is not precluded.

At most moments, a beam of unpolarized light has, of course, a sectional pattern that is elliptical. Hurwitz (H-41) has computed the average value of ellipticity, which turns out to be $\tan 15^\circ$ or 0.268.

No satisfactory way of describing unpolarized light pictorially has been found. To portray unpolarized light as a many-pointed star or asterisk is conventional, but without scientific merit; the portrayal fails to suggest the most prominent features of unpolarized light: its constantly changing, predominantly elliptical, character.

1.5. Partially Polarized Light. Since most light, whether of natural or artificial origin, is neither completely polarized nor completely unpolarized, the concept of *degree of polarization* is an important one. The quantity is defined in terms of the components into which the beam may be divided.

There are two useful ways of mentally dividing a beam into components. One method employs a “polarized–unpolarized” dichotomy, the other employs a “vibration-form” dichotomy. In employing the *polarized–unpolarized* dichotomy, one divides the given beam into a component C_a that is completely polarized and a component C_b that is unpolarized and has no long-term phase relation (no coherence) with C_a . It is a well-established fact that the outcome of the division process is unique: there is only one possible pair C_a and C_b . Depending on whether C_a is linearly, circularly, or elliptically polarized, the partially polarized beam in question is called partially linearly, partially circularly, or partially elliptically polarized.

In employing the *vibration-form* dichotomy, one divides the beam into that pair of completely and orthogonally polarized components that have the maximum difference in intensity. The more intense component may be called the *dominant component* C_d and the less intense component may be called the *inferior component* C_i . It may be shown that the two components are mutually incoherent; no long-term correlation between their phases exists.

The polarized–unpolarized dichotomy is useful in a variety of problems involving partial polarizers, retarders, and so forth. For example, if a partially polarized beam strikes a retarder, the unpolarized component is entirely unaffected thereby and accordingly the investigator is left free to concentrate his attention on the polarized component. The dichotomy has one notable limitation: no known device can perform the indicated analysis. (However, the converse process is easily achieved: it is an easy matter to combine a polarized beam and an unpolarized beam — locally, at least: one merely causes them to intersect at a small angle. By using this process, one can demonstrate the validity of the rationale under discussion.)

The vibration-form dichotomy is useful in problems involving those polarizers and polarizing beam splitters that employ birefringence. These devices are capable of accomplishing just such a dichotomy (within certain practical limitations).

The degree of polarization V is now easily defined. If the intensities

of components C_a , C_b , C_d , and C_i are called I_a , I_b , I_d , and I_i respectively, V is defined by either

$$V = \frac{I_a}{I_a + I_b}$$

or

$$V = \frac{I_d - I_i}{I_d + I_i}.$$

An author may wish to use a term that describes the incompleteness, or shortage, of polarization. The term *polarization defect*, defined as $I_i/(I_d + I_i)$, or its equivalent $(1 - V)/2$, may be used (S-11).

On some occasions, the concept of *equivalent degree of linear polarization* is a useful one. A beam that is 100-percent elliptically polarized and has a very small ellipticity might be judged to be not *completely, elliptically* polarized, but *incompletely, linearly* polarized; the naively arrived at result (equivalent degree of linear polarization) provides a figure that is useful in certain applications.

Spectrally variegated polarization is, of course, easy to achieve. Indeed, nearly any polychromatic beam emerging from a dichroic polarizer is likely to have a degree of polarization (and possibly even a polarization form) that varies with wavelength.

1.6. History of Understanding of Polarized Light. The history of man's understanding of polarized light has been touched on by Preston (P-30), Partington (P-10), and other authors. The early discoveries concerning double refraction are discussed in an anonymous work published in 1819 (A-10), and accounts of more recent developments are given by Archard (A-20), Johannsen (J-18), Thompson (T-5), Tutton (T-12), Twyman (T-13), and Wahlstrom (W-1). The development of the sheet-type polarizer has been described by Land and West (L-8) and by Land (L-13).

The following tabulation lists some of the principal advances:

- 1669 Erasmus Bartholinus, a Danish scientist, discovered double refraction (B-5).
- 1690 Christian Huyghens, a Dutch scientist, discovered polarization of light; he demonstrated the polarization with the aid of two calcite crystals arranged in series (H-42).
- 1757 Robert Hooke, an English physicist, tentatively suggested (perhaps before this year) that light vibrations are transverse (P-10).
- 1808 Etienne-Louise Malus, a French scientist, discovered polarization by

- reflection. He happened to be looking through a calcite crystal at the light reflected obliquely from a window of the Luxembourg Palace, in Paris, and observed that the two images produced by the calcite were extinguished alternately as he rotated the crystal (G-12; P-10; M-12).
- 1811 D. F. J. Arago, a French scientist, discovered optical rotation (P-10).
- 1812 Jean B. Biot, a French physicist, was the first to appreciate that refracto-uniaxial crystals could be classified according to whether the extraordinary index was greater or less than the ordinary index (J-18).
- 1812 Arago invented the pile-of-plates polarizer (A-18).
- 1812 David Brewster, a Scottish physicist, discovered "Brewster's law" regarding polarization by reflection (Ref. B-48 makes it clear that the date of discovery was 1812, not 1811 as maintained by some authors).
- 1815 Biot discovered tourmaline's peculiar property, dichroism (B-25).
- 1816 Augustin Fresnel, a French physicist, found that two rays that are polarized at right angles cannot interfere. But he was unable to explain the matter until enlightened by Young (F-15).
- 1817 Thomas Young, an English physicist, was the first to prove that light vibrations are transverse, as had been suggested by Hooke in 1757 or before (Y-2; P-10).
- 1828 William Nicol, a Scottish physicist, invented the nicol prism (J-18; L-8).
- 1844 Wilhelm Haidinger, an Austrian mineralogist, discovered the "Haidinger's brush" phenomenon by means of which one may perceive directly that a broad, uniform beam of linearly polarized light is indeed polarized (H-6).
- 1845 Michael Faraday, an English physicist, discovered the Faraday effect.
- 1847 Haidinger discovered circular dichroism (P-10; H-7).
- 1852 William B. Herapath, an English physician, discovered a synthetic crystalline material that polarizes light of all wavelengths in the visual range (H-23, H-24; L-13; G-12).
- 1852 George G. Stokes, an English physicist, invented the four "Stokes parameters" for describing a beam of partially polarized light (S-29).
- 1875 John Kerr, a Scottish physicist, discovered the Kerr effect (K-9).
- 1887 Heinrich Hertz, a German physicist, produced Hertzian waves, giving much support to Maxwell's theory.
- 1892 Henri Poincaré, a French mathematician, invented the "Poincaré-sphere" method of representing a beam of polarized light (P-20; R-1).
- 1928 Edwin H. Land, then a 19-year-old student at Harvard College, invented the first successful sheet-type dichroic polarizer (L-4, L-8).
- 1933 Bernard F. Lyot, a young French scientist, invented the polarization-type, narrow-band filter that now bears his name (L-28).
- 1938 Edwin H. Land invented H-sheet, a dichroic sheet-type polarizer

that relies on a polymeric molecular type of absorber rather than a crystalline absorber.

- 1940 Robert Clark Jones, an American physicist, invented the Jones calculus for computing the changes produced in beams of light by polarizers and retarders (J-19).
- 1942 Francis Perrin, a French scientist, found how to put the four Stokes parameters and the sixteen Soleillet transformation constants into compatible form — a form involving matrix algebra (P-12).
- 1943 Hans Mueller, a professor of physics at Massachusetts Institute of Technology, invented in this year, or shortly before, a phenomenological approach to problems involving polarized and partially polarized light; the approach made extensive use of 4×4 matrices (M-26, M-28).
- 1943 Robert P. Blake, an American scientist, invented the HR-type polarizer, the first sheet-type polarizer suitable for use throughout an appreciable part of the infrared range (B-31; U.S. patent 2,494,686).

2.1. Introduction. We now turn to the modern methods of describing polarized light: the Poincaré sphere, the Stokes vector, the Jones vector, and the quantum-mechanical representation. We give particular attention to the Poincaré sphere and the Stokes vector because they provide direct insight into certain difficult problems and permit great simplification in various calculations of what happens to a beam when it encounters a succession of polarizers and retarders.

2.2. Poincaré Sphere. The Poincaré sphere, conceived by Henri Poincaré in about 1892 (P-20), provides a convenient way of representing polarized light and predicting how any given retarder will change the polarization form. The method is essentially one of mapping: each point on the sphere represents a different polarization form. The mapping may be carried out with the aid of a three-dimensional model, a two-dimensional projection, trigonometry, or analytic geometry. (The method is applicable only if the incident beam is completely polarized.)

In addition to its mapping function, the sphere leads to short-cut solutions of problems involving retarders, or combinations of retarders and ideal homogeneous polarizers. With the sphere in mind, one may often see at a glance the solution of a problem that would be difficult to solve by conventional methods. This is true, for example, of problems encountered by Pancharatnam (P-1, P-2) and Koester (K-12) in searching for a design of an achromatic, multilayer, retarder.

Despite its great usefulness, the Poincaré sphere has been given little

attention in the literature. The writer has been unable to find any substantial account of it in any textbook. However, brief descriptions have appeared in a number of technical journals; see Ramachandran and Ramaseshan (R-1, R-2), Jerrard (J-14), Koester (K-12), McMaster (M-2), Perrin (P-12), Skinner (S-17), Walker (W-3), and Wright (W-28).

Figure 2.1 indicates the significance of the various parts of the sphere. The upper and lower poles represent left- and right-circularly

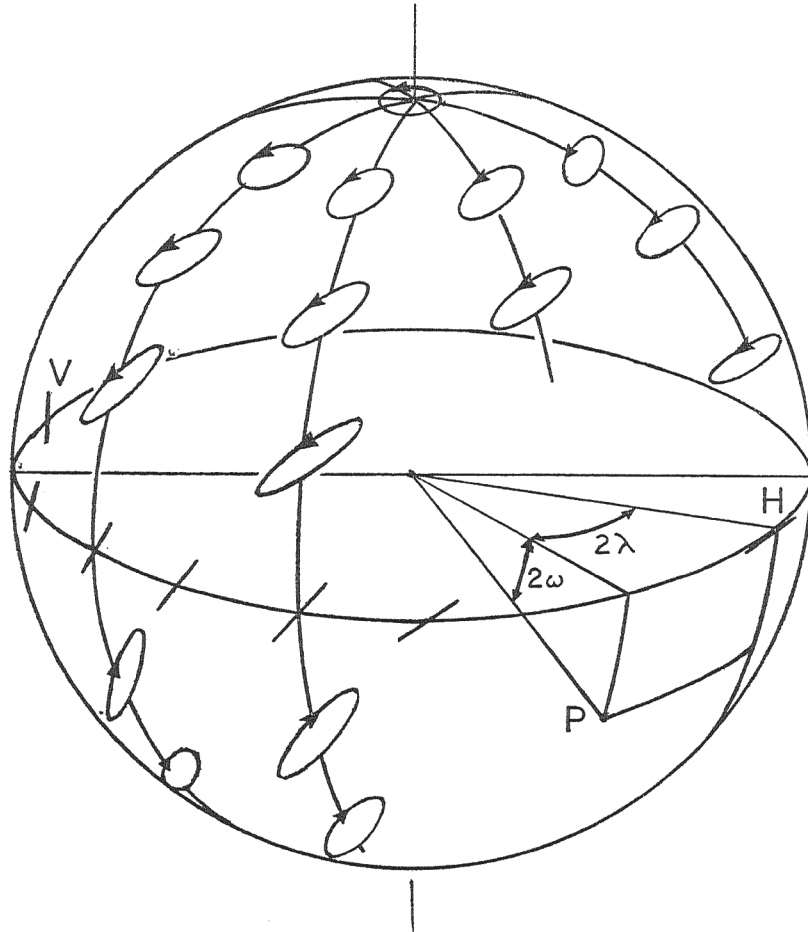


FIG. 2.1. The Poincaré sphere, showing the general significance of the different areas and the specification of the general point P in terms of the angles 2λ , measured clockwise from H , and 2ω , measured downward from the equator.

polarized light. Points on the equator indicate linear polarization. Other points indicate elliptical polarization. An arbitrarily chosen point H on the equator designates horizontal polarization, and the diametrically opposite point V designates vertical polarization. Beams whose polarization forms are orthogonal are represented by points at opposite ends of a diameter. Usually the radius of the sphere is assumed to be unity; sometimes it is regarded as proportional to the intensity of the beam.

A general point P on the surface of the (unit-radius) Poincaré sphere is specified in terms of the longitude (2λ) and the latitude (2ω), where $-180^\circ \leq 2\lambda \leq 180^\circ$ and $-90^\circ \leq 2\omega \leq 90^\circ$. (Note that the symbols λ and ω have quite different meanings from those implied in the previous chapter.) The longitude is positive when measured clockwise from point H ; the latitude is positive when measured downward from the equator, that is, toward the pole representing right circularly polarized light. Thus the coordinates of the point P in Fig. 2.1 are positive.

The significance of the general point P is easily stated. It represents a completely polarized beam whose ellipse has azimuth λ , ellipticity $\tan |\omega|$, and a handedness that is left or right according to whether P lies in the upper or lower hemisphere. In summary, P represents an elliptically polarized beam whose sectional pattern is described thus:

$$\begin{aligned}\alpha &= \lambda, \\ b/a &= \tan |\omega|,\end{aligned}$$

handedness: left or right for 2ω negative or positive respectively.

Obviously, each point on the sphere represents a different polarization form. Conversely, each polarization form is represented by a different point on the sphere.

Cartesian Coordinates. Alternatively, P may be specified by means of the right-handed cartesian coordinates X , Y , and Z indicated in Fig. 2.2. (These coordinates are not to be confused with those used for a different purpose in Chapter 1.) When the sphere has unit radius, the coordinates of P are:

$$\begin{aligned}X &= \cos 2\omega \cos 2\lambda, \\ Y &= \cos 2\omega \sin 2\lambda, \\ Z &= \sin 2\omega.\end{aligned}$$

When the light is horizontally polarized, $2\omega = 2\lambda = 0$, and the full expressions reduce to $X = 1$, $Y = 0$, $Z = 0$. When the light is

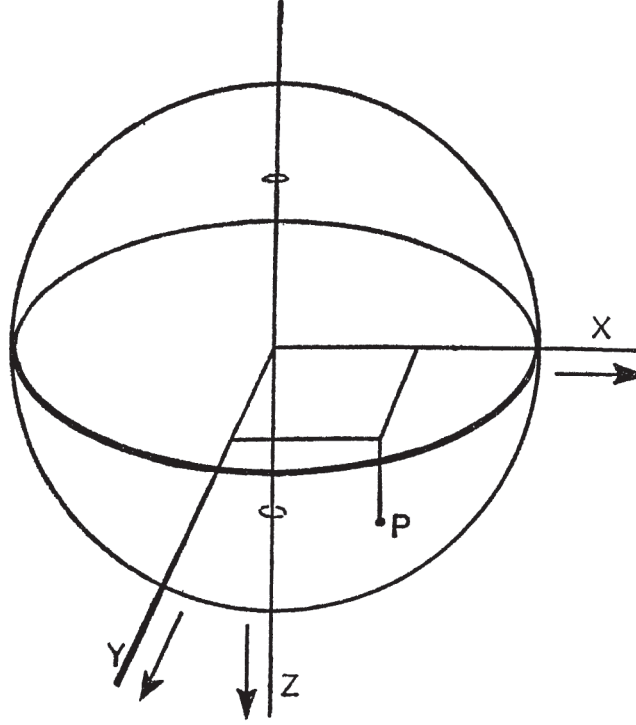


FIG. 2.2. Axes of the cartesian coordinate system used in specifying a point P on the unit-radius Poincaré sphere. Note that the positive Z -axis extends *downward*.

vertically polarized, $2\omega = 0$, $2\lambda = \pi$, and we find that $X = -1$, $Y = 0$, $Z = 0$. (We shall see in the next section that, in general, any completely polarized beam has X , Y , and Z values that are identical to the normalized Stokes parameters M , C , and S respectively. Thus the M , C , and S values shown in Table 2.1 are applicable here also.)

From the equations presented above, one sees that if the X , Y , and Z values of a beam are given, the azimuth, ellipticity, and handedness may be computed from the formulas

$$\alpha = \lambda = \frac{1}{2} \arctan \frac{Y}{X},$$

$$\frac{b}{a} = \tan |\omega| = \tan \left| \frac{1}{2} \arcsin Z \right|,$$

handedness: left or right for Z negative or positive respectively.

The main use of the Poincaré sphere is in finding the effect of inserting a retarder in a polarized beam. The effect may be found merely by constructing an arc of a circle on the sphere. The method is described in detail in Chapter 7, and only a brief summary is needed here. The choice of point (on the sphere) to serve as center of the arc depends solely on the type of retarder (its fast eigenvector's azimuth, ellipticity, and handedness); the choice of arc length depends solely on the retardance. One end of the arc is defined by the point that describes the incident beam. The other end provides the answer: its location describes the emerging beam. Chapter 7, besides describing the procedure in detail, presents specific applications.

In general, the Poincaré-sphere representation is useful not merely in *solving* problems but also in *formulating* them — it clarifies one's thinking and talking. It is a kind of blackboard on which to indicate the starting point, the procedure, and the outcome, all with maximum economy of effort. The principal step, obviously, is drawing an arc or, to use a more common expression, *rotating the sphere*.

2.3. Stokes Vector. Though conceived in 1852 (S-29), the Stokes vector has received but scant attention in the literature of optics. Perhaps the best of the modern accounts is that by Walker (W-3). Other pertinent articles are those by Soleillet (S-24), Perrin (P-12), Billings and Land (B-22), Ramachandran and Ramaseshan (R-1), McMaster (M-2, M-4), Tinkham and Strandberg (T-7), Pancharatnam (P-4), Falkoff and McDonald (F-3). Various articles by R. Clark Jones, by H. Mueller, and by S. Chandrasekhar may be cited also; see bibliography. Van de Hulst's 1957 book (V-1) applies the Stokes vector to the problem of light scattering; Cohen's 1958 article (C-18) applies it to radio astronomy.

The Stokes vector consists of a set of four quantities (called the Stokes parameters) that describe the intensity and polarization of a beam of light. The beam may be polarized completely, partially, or not at all; it may be monochromatic or polychromatic. Thus the description, though very concise, is broadly applicable.

The four parameters have the dimensions of intensity; each corresponds not to an instantaneous intensity but to a time-averaged intensity, the average being taken over a period long enough to permit practical measurement. The vector, though consisting of four physically real parameters, is, of course, a *mathematical* vector; it exists in

a four-dimensional mathematical space, not in a three-dimensional physical space.

Following the practice adopted by Perrin and by Jones (P-12; J-19), we shall call the four quantities I , M , C , and S . (Stokes himself used A , B , C , D ; Walker used I , Q , U , V). The four quantities comprise a column vector:

$$\begin{bmatrix} I \\ M \\ C \\ S \end{bmatrix}.$$

The vector is often written horizontally, to save space; curly brackets are then used, as a reminder that the quantity is indeed a column vector:

$$\{I, M, C, S\}.$$

The importance of writing the quantities in an invariant order is made clear in Chapter 8, where the use of the vector in conjunction with 4×4 Mueller matrices is discussed.

The first parameter, I , is called the intensity. The parameters M , C , and S may be regarded as the "horizontal preference," "plus 45° preference," and "right circular preference" respectively. When a parameter has a negative value, the preference is for the orthogonal polarization form; thus if the parameter S has the value -0.5 the polarization form is more akin to left-circular polarization than to right-circular polarization.

Definitions. Although the parameters can be defined in terms of the electromagnetic theory, we prefer to define them operationally. This approach is perhaps more basic and reliable; also, it corresponds more closely to the historical development (S-29).

The operational approach makes use of a set of four filters. Any of several different sets could be used, but to make our account specific we shall assume a set of four filters F_1 , F_2 , F_3 , F_4 that have the following properties: each has a transmittance of 0.5 for incident, unpolarized light; each is oriented so that its faces are vertical and are perpendicular to the beam; F_1 has the same effect on any incident beam, irrespective of the beam's polarization; in short, it is isotropic; F_2 is opaque to incident light that is polarized with its electric vibration direction at 90° , that is, vertical; F_3 is opaque to light polarized at -45° ; F_4 is opaque to left-circularly polarized light.

Obviously the filters represent nonpolarizing, linearly (horizontally) polarizing, linearly (plus 45°) polarizing, and right-circularly polarizing filters respectively, as implied by Fig. 2.3.

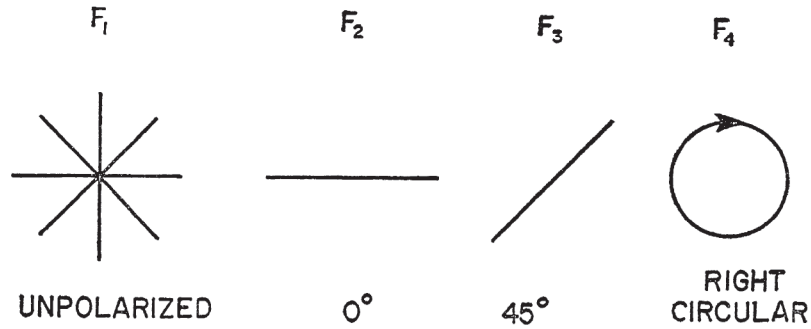


FIG. 2.3. Polarization forms produced by the four filters used in determining the four Stokes parameters of a beam.

In addition to the filters, a detector is employed. It is of polarization-independent type, and is calibrated in terms of intensity, or, more specifically, power. Its cross-sectional area must, of course, be at least as great as that of the beam in question.

The procedure is to place the detector squarely in the beam, interpose each of the four filters successively, note the four values indicated by the detector, and multiply each value by $1/0.5$. From the resulting four values, which may be called V_1 , V_2 , V_3 , and V_4 (and are not to be confused with the symbol that stands for degree of polarization!), we calculate the four parameters, thus:

$$\begin{aligned}
 I &= V_1, \\
 M &= V_2 - V_1, \\
 C &= V_3 - V_1, \\
 S &= V_4 - V_1.
 \end{aligned}$$

The method of defining the parameters in terms of electromagnetic theory has been indicated by Walker (W-3). It is awkward in that one must assume that the light is sufficiently monochromatic that, at any time, a definable phase angle γ exists between the instantaneous scalar components a_x and a_y of the electric field, yet the light must be sufficiently *polychromatic* that the unpolarized state is not precluded. (The angle γ is defined so that, when $\pi > \gamma > 0$, the handedness of polarization is right; when $-\pi < \gamma < 0$ the handedness is left.)

The definitions are as follows (the angular brackets are employed to indicate that *time averages* are meant):

$$\begin{aligned} I &= \langle a_x^2 + a_y^2 \rangle, \\ M &= \langle a_x^2 - a_y^2 \rangle, \\ C &= \langle 2a_x a_y \cos \gamma \rangle, \\ S &= \langle 2a_x a_y \sin \gamma \rangle. \end{aligned}$$

As before, the four quantities have the dimensions of intensity. As before, M , C , and S may be positive, negative, or zero.

Normalized Parameters. Often we are interested only in relative values. We then divide all the parameters by the first one, to obtain the *normalized parameters*. Thus an original vector $\{4, 0, 2, 0\}$ becomes the normalized vector $\{1, 0, 0.5, 0\}$.

Typical Vectors. It is a simple exercise to compute, from the definitions based on electromagnetic theory, various vectors of principal interest. Consider, for example, unpolarized light. There is no time-averaged preference between a_x and a_y ; consequently $\langle a_x^2 + a_y^2 \rangle$ reduces to $\langle 2a_x^2 \rangle$ and $\langle a_x^2 - a_y^2 \rangle$ reduces to zero. The multiplicative factor $\cos \gamma$ has a sign and magnitude that are independent of the sign and magnitude of the product $a_x a_y$; thus $\langle a_x a_y \cos \gamma \rangle$ reduces to zero. The same applies to $\langle a_x a_y \sin \gamma \rangle$. Consequently unpolarized light is described by an original vector $\{2a_x^2, 0, 0, 0\}$ and by the normalized vector $\{1, 0, 0, 0\}$.

Similar reasoning shows that a horizontally polarized beam has the original vector $\{a_x^2, a_x^2, 0, 0\}$ and the normalized vector $\{1, 1, 0, 0\}$.

With little effort one may compute the Stokes vectors for many different forms of completely polarized light. Table 2.1 lists these vectors.

None of the parameters can be larger than the first (I). Each of the others lies in the range from $-I$ to $+I$. If the beam is entirely unpolarized, $M = C = S = 0$. If it is completely polarized, $(M^2 + C^2 + S^2)^{1/2} = I$. If the degree of polarization is V , it follows that $(M^2 + C^2 + S^2)^{1/2}/I = V$.

The parameter M is positive if the polarization form is more akin to a horizontal line than to a vertical line; it is negative if the preference is for a vertical line; it has the value zero when there is no preference between these, as for example when the polarization form is circular, or corresponds to an ellipse whose major axis is at $\pm 45^\circ$. The parameter C is positive or negative depending on whether the

Table 2.1. Stokes vectors and Jones vectors of various forms of polarized light.

Sectional pattern	Polarization form				Normalized Stokes vector,				Jones vector	
	α (deg)	b/a	A_y/A_x	γ (deg)	I	M	C	S	Standard normalized	Full
	0	0	0	—	{1, 1, 0, 0}				$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ 0 \end{bmatrix}$
	90	0	∞	—	{1, -1, 0, 0}				$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 0 \\ A_y e^{i\epsilon_y} \end{bmatrix}$
	45	0	1	0	{1, 0, 1, 0}				$\frac{\sqrt{2}}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ A_x e^{i\epsilon_x} \end{bmatrix}$
	-45	0	1	± 180	{1, 0, -1, 0}				$\frac{\sqrt{2}}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ -A_x e^{i\epsilon_x} \end{bmatrix}$
General	linear	0	Any positive number	0 or ± 180	{1, $\cos 2\alpha$, $\sin 2\alpha$, 0}				$\begin{bmatrix} \cos R \\ \pm \sin R \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ \pm A_y e^{i\epsilon_y} \end{bmatrix}$
	—	$1, R$	1	90	{1, 0, 0, 1}				$\frac{\sqrt{2}}{2} \begin{bmatrix} -i \\ 1 \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ A_x e^{i(\epsilon_x + \pi/2)} \end{bmatrix}$
	—	$1, L$	1	-90	{1, 0, 0, -1}				$\frac{\sqrt{2}}{2} \begin{bmatrix} i \\ 1 \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ A_x e^{i(\epsilon_x - \pi/2)} \end{bmatrix}$
	0	$\frac{1}{2}, R$	$\frac{1}{2}$	90	{1, 0.6, 0, 0.8}				$\frac{2\sqrt{5}}{5} \begin{bmatrix} -i \\ 1 \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ \frac{1}{2} A_x e^{i(\epsilon_x + \frac{\pi}{2})} \end{bmatrix}$
	90	$\frac{1}{2}, R$	2	90	{1, -0.6, 0, 0.8}				$\frac{2\sqrt{5}}{5} \begin{bmatrix} -i \\ 1 \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ 2 A_x e^{i(\epsilon_x + \frac{\pi}{2})} \end{bmatrix}$
	22.5	$0.318, 0.518, R$		45	$\left\{1, \sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}}\right\}$				$0.325 \begin{bmatrix} 2.73 \\ 1+i \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ 0.518 A_x e^{i(\epsilon_x + \frac{\pi}{4})} \end{bmatrix}$
General elliptical					$\begin{bmatrix} 1 \\ \cos 2\omega \cos 2\lambda \\ \cos 2\omega \sin 2\lambda \\ \sin 2\omega \end{bmatrix}$				$\begin{bmatrix} (\cos R) e^{-i\frac{\gamma}{2}} \\ (\sin R) e^{i\frac{\gamma}{2}} \end{bmatrix}$	$\begin{bmatrix} A_x e^{i\epsilon_x} \\ A_y e^{i\epsilon_y} \end{bmatrix}$
General elliptical, partially polarized					$\frac{1}{\langle a_x^2 + a_y^2 \rangle} \begin{bmatrix} \langle a_x^2 + a_y^2 \rangle \\ \langle a_x^2 - a_y^2 \rangle \\ \langle 2a_x a_y \cos \gamma \rangle \\ \langle 2a_x a_y \sin \gamma \rangle \end{bmatrix}$				None	None
Unpolarized					{1, 0, 0, 0}				None	None

polarization form is more akin to the $+45^\circ$ direction than to the -45° direction. The parameter S is positive for right-handed polarization forms, negative for left-handed forms, and zero for all linear forms.

Unit-intensity beams that are orthogonally polarized have Stokes vectors that differ only in the signs of the second, third, and fourth parameters; for example, the beams $\{1, 0.6, 0, 0.8\}$ and $\{1, -0.6, 0, -0.8\}$ are orthogonally polarized.

Principle of Optical Equivalence. After defining his vector, Stokes announced his “principle of optical equivalence” (S-29). According to this principle, beams that have the same Stokes vector are indistinguishable as regards intensity, degree of polarization, and polarization form.

(The principle is sometimes worded so as to imply that two beams having the same Stokes vector are indistinguishable. Such a statement is inaccurate; indeed, it is inaccurate even in cases where the two beams have, in addition, the same spectral energy distribution and the same geometry. For the fact is that the two beams will differ, in general, as regards phase, or fluctuations in phase; and by using an interferometric method such as that of Langsdorf and DuBridge (L-14) an investigator may be able to distinguish between the two beams.)

Types of Applications. The simplest application of the Stokes vector is to the combining of beams. When a beam 1 is combined with a beam 2 — and the two beams are *incoherent* — the properties of the combined beam c are found by adding the two vectors:

$$\begin{aligned} I_c &= I_1 + I_2, \\ M_c &= M_1 + M_2, \\ C_c &= C_1 + C_2, \\ S_c &= S_1 + S_2. \end{aligned}$$

Example: If the two initial beams have the vectors $\{3, 1, 1, -2\}$ and $\{4, 0, 0, 4\}$, the combined vector is $\{7, 1, 1, 2\}$.

Since the different areas of most light sources are incoherent with respect to one another, the Stokes vector is entirely applicable to the adding of beams originating at different areas of the source. (If the beams are coherent, or partially coherent, the procedure is not applicable, or is applicable only when certain precautions are taken. Pancharatnam, for example, has succeeded in using the Stokes vector in dealing with the interference of partially coherent beams; see

bibliography. An investigator who is concerned with coherent beams will usually prefer the Jones calculus.)

Perhaps the most important application of the Stokes vector is in connection with the Mueller calculus (Chapter 8), which affords a superb mathematical framework for computing the change produced in the intensity and polarization of a beam when it encounters a train of polarizers, retarders, and scatterers.

The Stokes vector of a completely polarized beam is closely related to the Poincaré-sphere representation. Consider the four quantities:

$$\begin{aligned} I &= a_x^2 + a_y^2, \\ M &= a_x^2 - a_y^2, \\ C &= 2a_x a_y \cos \gamma, \\ S &= 2a_x a_y \sin \gamma. \end{aligned}$$

As we have seen, $I^2 = M^2 + C^2 + S^2$. Consequently the first quantity may be regarded as the radius of a sphere and the other three may be regarded as the cartesian coordinates of a point on the sphere. The cartesian axis corresponding to the second quantity, $(a_x^2 - a_y^2)$, that is, M , is to be associated with horizontal polarization, since, for a given I , the quantity $(a_x^2 - a_y^2)$ is maximized when $a_y = 0$. Similarly the axes corresponding to the last two quantities are the Y - and Z -axes. In summary, the M , C , and S values of a completely polarized beam correspond to the X , Y , and Z values of the Poincaré-sphere representation of that beam.

2.4. Jones Vector. The Jones vector, introduced in 1941 by R. Clark Jones (J-19), describes a polarized beam with the maximum algebraic brevity, and is eminently suited to the solving of problems involving beams whose phase relations are important. The present section defines the Jones vector, lists its commonest forms, and relates it to the more familiar ways of describing polarized light. Also, it shows how the vector is used in problems involving the combining of polarized beams of any polarization form and any phase relationship. (The use of the vector in problems involving polarizers and retarders is treated in Chapter 8.)

The Jones vector is a two-element column vector which describes a beam's polarization form and amplitude components at some given position along the beam. If the beam is traveling along the Z -axis, the vector has the general form:

$$\begin{bmatrix} m \\ n \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} E_x \\ E_y \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} A_x e^{i(\epsilon_x + 2\pi\nu t)} \\ A_y e^{i(\epsilon_y + 2\pi\nu t)} \end{bmatrix},$$

where E_x and E_y are the scalar components (of the instantaneous electric vector) along the X - and Y -axes; A_x is the maximum value of E_x , and A_y is the maximum value of E_y . The quantity ϵ_x is the phase of the component E_x at time $t = 0$ and at the given location; ϵ_y is the phase of the component E_y . (As in Chapter 1, the letter R may be used to represent $\arctan A_y/A_x$.) In general, each element of the column vector is a complex quantity.

We may convert the vector to the following equivalent form:

$$e^{i2\pi\nu t} \begin{bmatrix} A_x e^{i\epsilon_x} \\ A_y e^{i\epsilon_y} \end{bmatrix}.$$

Since the magnitude of any quantity of the form e^{iM} is unity, the magnitude of $e^{i2\pi\nu t}$ is unity. Thus this latter quantity may be dropped entirely in those problems in which details of variation with time are not of interest. Most problems are of this type, and accordingly the Jones vector is often written in the following form, called the *full Jones vector*:

$$\begin{bmatrix} A_x e^{i\epsilon_x} \\ A_y e^{i\epsilon_y} \end{bmatrix}.$$

Obviously, if it turns out that the time variation is of interest, the factor $e^{i2\pi\nu t}$ can be restored at any desired stage of the calculation.

In certain cases the full vector can be simplified further. Consider horizontally polarized light: here $A_y = 0$, so the vector reduces to

$$\begin{bmatrix} A_x e^{i\epsilon_x} \\ 0 \end{bmatrix}.$$

For light that is linearly polarized at 45° , the two quantities A_x and A_y are equal, the quantities ϵ_x and ϵ_y are equal, and the vector reduces to

$$\begin{bmatrix} A_x e^{i\epsilon_x} \\ A_x e^{i\epsilon_x} \end{bmatrix} \quad \text{or} \quad A_x e^{i\epsilon_x} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For right-circularly polarized light, $A_x = A_y$ and $\gamma \equiv \epsilon_y - \epsilon_x = \frac{1}{2}\pi$. Thus the vector is

$$\begin{bmatrix} A_x e^{i\epsilon_x} \\ A_x e^{i(\epsilon_x + \frac{1}{2}\pi)} \end{bmatrix}.$$

Table 2.1 lists the full Jones vectors for various forms of polarization.

Intensity of the Beam. The intensity of the beam is proportional to the sum of the squares of the magnitudes of the individual elements. If the units of intensity or amplitude are chosen so that the proportionality constant is unity, the relation becomes

$$I = A_x^2 + A_y^2.$$

Standard Normalized Jones Vector. Any full vector may be converted to standard normalized form by multiplying it by whatever scalar (ordinarily a complex scalar) reduces the intensity to unity and also reduces the vector to simplest form. The process is called normalizing. Examples of nonnormalized and normalized vectors are as follows:

<i>Nonnormalized Form</i>	<i>Standard Normalized Form</i>
$\begin{bmatrix} 3e^{i\pi} \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\frac{\sqrt{2}}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
$\frac{1}{2} \begin{bmatrix} 1+i \\ 1-i \end{bmatrix}$	$\frac{\sqrt{2}}{2} \begin{bmatrix} i \\ 1 \end{bmatrix}$

Table 2.1 lists the standard normalized vectors for various forms of polarized light.

If a normalized Jones vector is denoted by $\begin{bmatrix} m \\ n \end{bmatrix}$, the vector of an orthogonally polarized beam is $\begin{bmatrix} -n^* \\ m^* \end{bmatrix}$, where the asterisk denotes the complex conjugate. Thus, for example, the following two forms are orthogonal:

$$\frac{\sqrt{2}}{2} \begin{bmatrix} -i \\ 1 \end{bmatrix} \quad \text{and} \quad \frac{\sqrt{2}}{2} \begin{bmatrix} -1 \\ i \end{bmatrix}.$$

Interpretation of the Vector. The polarization form implied by a given Jones vector may be expressed in conventional terms with the aid of the following "tabulation of inferences":

<i>Status of E_x and E_y</i>	<i>Inference as to Polarization Form</i>
$E_x \neq 0, E_y = 0$	Linear; $\alpha = 0^\circ$
$E_x = 0, E_y \neq 0$	Linear; $\alpha = 90^\circ$
$E_x \neq 0, E_y \neq 0, E_x = E_y$	Linear; $\alpha = 45^\circ$
$E_x \neq 0, E_y \neq 0, E_x = -E_y$	Linear; $\alpha = -45^\circ$
$E_y/E_x = i$	Right circular
$E_y/E_x = -i$	Left circular
General case	Elliptical; see below

When a vector not falling in one of the simple cases is encountered, a three-step procedure is followed. Step 1 consists in converting the vector to the form of the full Jones vector:

$$\begin{bmatrix} A_x e^{i\epsilon_x} \\ A_y e^{i\epsilon_y} \end{bmatrix}.$$

Thus if an investigator encounters the vector

$$\begin{bmatrix} 0.3i \\ 0.5e^{i(-0.2)} \end{bmatrix},$$

he alters the first element by replacing i by its equivalent $e^{i\pi/2}$, and obtains:

$$\begin{bmatrix} 0.3e^{i\pi/2} \\ 0.5e^{i(-0.2)} \end{bmatrix}.$$

Step 2 consists in computing the angles R and γ , defined in the previous chapter as $|\arctan(A_y/A_x)|$ and $(\epsilon_y - \epsilon_x)$. In the present example,

$$R = |\arctan(0.5/0.3)| = 59.0^\circ,$$

$$\gamma = (-0.2 - \tfrac{1}{2}\pi) = -1.77 \text{ rad or } -101^\circ.$$

Step 3 consists in employing the following rules:

If $\sin \gamma > 0$, the ellipse is right-handed; if $\sin \gamma < 0$, the ellipse is left-handed;

The azimuth α of the major axis is given by

$$\alpha = \tfrac{1}{2} \arctan [(\tan 2R) (\cos \gamma)];$$

The ellipticity b/a is given by

$$b/a = \tan \beta,$$

where $\beta = \tfrac{1}{2} \arcsin (\sin 2R |\sin \gamma|)$.

Applying these rules to the present example, one finds that:

$\sin \gamma < 0$, and consequently the ellipse is left-handed;

$$\begin{aligned}\alpha &= \frac{1}{2} \arctan [\tan 118^\circ \cos (-101^\circ)] \\ &= \frac{1}{2} \arctan 0.37 = 10^\circ, \\ \beta &= \frac{1}{2} \arcsin [\sin 118^\circ |\sin (-101^\circ)|] \\ &= \frac{1}{2} \arcsin 0.86 = 30^\circ, \\ b/a &= \tan 30^\circ = 0.58.\end{aligned}$$

An investigator who is familiar with the Jones vector can easily compute the result of combining two coherent beams of completely polarized light. He writes down the two full vectors appropriate to the given polarization forms, given intensities, and given phases. He then adds the vectors.

Consider, for example, the result of adding two linearly polarized beams, one of which is polarized horizontally and the other vertically. Let each beam have an intensity of 3 units, and let the vertically polarized beam be 90° ahead in phase. The two full vectors are:

$$\begin{bmatrix} \sqrt{3}e^{i\epsilon_z} \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 \\ \sqrt{3}e^{i(\epsilon_z+90^\circ)} \end{bmatrix}.$$

The sum is

$$\begin{bmatrix} \sqrt{3}e^{i\epsilon_z} \\ \sqrt{3}e^{i(\epsilon_z+90^\circ)} \end{bmatrix} = \sqrt{3}e^{i(\epsilon_z+90^\circ)} \begin{bmatrix} e^{-i(90^\circ)} \\ 1 \end{bmatrix} = \sqrt{3}e^{i(\epsilon_z+90^\circ)} \begin{bmatrix} -i \\ 1 \end{bmatrix}.$$

Referring to Table 2.1 one sees that this last expression represents a beam that is right-circularly polarized and has an intensity of $2(\sqrt{3})^2$ or 6.

The most important applications of the Jones vector are in connection with the Jones calculus (Chapter 8), which is the most succinct of the algebraic methods of computing the outcome of passing a completely polarized beam through a series of ideal retarders and polarizers. Other applications are discussed by Ramachandran and Ramaseshan (R-2).

2.5. Quantum-mechanical Description. The great majority of problems involving polarizers and polarized light are solved without reference to quantum mechanics. However, the relation between the quantum-mechanical specification of light and the more conventional specifications (presented in the foregoing sections) is of interest. The relation has been discussed by Jauch and Rohrlich (J-6), Fano (F-5, F-6), and McMaster (M-5).

According to the quantum-mechanical viewpoint, a given beam of completely polarized, monochromatic light consists of photons that are identical. Each is in the identical state (a pure state), and has the same wave function. This is true not only of a linearly polarized beam, but also of a circularly or elliptically polarized beam; the wave function that applies is different, of course, for each polarization type. In general, no one polarization type is more basic, or simpler, than any other; however, if one is dealing mainly with spin operators, right- and left-circularly polarized photons are specified with maximum simplicity.

Even an *unpolarized* (monochromatic) beam consists, according to the quantum-mechanical viewpoint, of identical photons, and each of these has the same wave function. However, the wave function represents a mixed state, that is, an incoherent combination, described by an expression of the form

$$|a\psi_1| + |b\psi_2|,$$

whereas the specification of the general, pure state (representing the general case of elliptical polarization and constituting a coherent combination) has the form

$$|a\psi_1 + b\psi_2|.$$

The statement that all the photons of the unpolarized beam are identical may at first seem surprising, but is in accord with the important experimental fact that, a priori, our expectation concerning the behavior of any one photon (from the given unpolarized beam) is the same as the expectation concerning any other photon of this beam. It is in accord, in short, with the fact that an unpolarized beam contains no information as to its history, that is, as to whether it was created by the incoherent combination of two orthogonally linearly polarized forms, two orthogonal circular forms, two orthogonal elliptical forms, or in some more complicated way.

The Stokes vector, also, takes no account of the history of the given beam but depends solely on expectation probabilities. Accordingly, its relation to the quantum-mechanical description of light is a close one. Indeed, the Stokes vector serves as a kind of bridge between the classical theory of light and the quantum mechanical theory; see, for example, McMaster's monographs M-2 and M-5.

2.6. Thermodynamics and Entropy. A completely polarized, nearly monochromatic ray of given intensity can be assigned a temperature

and also a value of entropy flux, as explained by Planck (P-17, P-18). If the beam is only partially polarized, two temperatures are involved, that is, a different temperature must be assigned to the dominant component and the inferior component, defined in Sec. 1.5. Likewise two values of entropy flux are involved. If the beam is entirely unpolarized, the two temperatures are identical, and the two values of entropy flux are identical. (If it is completely polarized, we may still say that two temperatures exist, and that one of the temperatures is zero; likewise we may say that two entropy fluxes exist, and that one of these is zero.)

A 1-watt beam of 500-m μ light that is completely polarized has a temperature that is higher than that of either component of a 1-watt, 500-m μ beam that is unpolarized (and has the same width and angular spread); the entropy flux of the polarized beam is less than the sum of the entropy fluxes of the two components of the unpolarized beam. Of course, the temperature and the entropy flux of a completely polarized beam are independent of the polarization form concerned.

For additional information, see Born (B-41), Jones (J-30), Ore (O-3), and Rosen (R-12).

The relation of polarized light and general relativity has been discussed by Mariot (M-14).